

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z1,Z2,Z5

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z1 Z2 Z5: $\frac{Z_1Z_2Z_5g_m-Z_1Z_2+Z_1Z_5}{2Z_1Z_2g_m+4Z_1+Z_2+Z_5}$

$$H(z) = \frac{Z_1Z_2Z_5g_m - Z_1Z_2 + Z_1Z_5}{2Z_1Z_2g_m + 4Z_1 + Z_2 + Z_5}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1, \frac{1}{C_2s}, \infty, \infty, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{R_1R_5g_m - R_1 + s(C_2R_1R_5 - C_5R_1R_5)}{4C_2C_5R_1R_5s^2 + 2R_1g_m + s(4C_2R_1 + C_2R_5 + 2C_5R_1R_5g_m + C_5R_5) + 1}$$

Parameters:

Q: $\frac{2\sqrt{C_2}\sqrt{C_5}\sqrt{R_1}\sqrt{R_5}\sqrt{2R_1g_m+1}}{4C_2R_1+C_2R_5+2C_5R_1R_5g_m+C_5R_5}$
wo: $\frac{\sqrt{2R_1g_m+1}}{2\sqrt{C_2}\sqrt{C_5}\sqrt{R_1}\sqrt{R_5}}$
bandwidth: $\frac{4C_2R_1+C_2R_5+2C_5R_1R_5g_m+C_5R_5}{4C_2C_5R_1R_5}$
K-LP: $\frac{R_1R_5g_m-R_1}{2R_1g_m+1}$
K-HP: 0
K-BP: $\frac{C_2R_1R_5-C_5R_1R_5}{4C_2R_1+C_2R_5+2C_5R_1R_5g_m+C_5R_5}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(R_1, \frac{R_2}{C_2R_2s+1}, \infty, \infty, \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{R_1R_2g_m + R_1 + s(C_2R_1R_2 - C_5R_1R_2)}{4C_2C_5R_1R_2s^2 + s(C_2R_2 + 2C_5R_1R_2g_m + 4C_5R_1 + C_5R_2) + 1}$$

Parameters:

Q: $\frac{2\sqrt{C_2}\sqrt{C_5}\sqrt{R_1}\sqrt{R_2}}{C_2R_2+2C_5R_1R_2g_m+4C_5R_1+C_5R_2}$
wo: $\frac{1}{2\sqrt{C_2}\sqrt{C_5}\sqrt{R_1}\sqrt{R_2}}$
bandwidth: $\frac{C_2R_2+2C_5R_1R_2g_m+4C_5R_1+C_5R_2}{4C_2C_5R_1R_2}$
K-LP: $R_1R_2g_m + R_1$
K-HP: 0
K-BP: $\frac{C_2R_1R_2-C_5R_1R_2}{C_2R_2+2C_5R_1R_2g_m+4C_5R_1+C_5R_2}$
Qz: None
Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(R_1, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{R_1 R_2 R_5 g_m - R_1 R_2 + R_1 R_5 + s (C_2 R_1 R_2 R_5 - C_5 R_1 R_2 R_5)}{4 C_2 C_5 R_1 R_2 R_5 s^2 + 2 R_1 R_2 g_m + 4 R_1 + R_2 + R_5 + s (4 C_2 R_1 R_2 + C_2 R_2 R_5 + 2 C_5 R_1 R_2 R_5 g_m + 4 C_5 R_1 R_5 + C_5 R_2 R_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_2}\sqrt{C_5}\sqrt{R_1}\sqrt{R_2}\sqrt{R_5}\sqrt{2R_1R_2g_m+4R_1+R_2+R_5}}{4C_2R_1R_2+C_2R_2R_5+2C_5R_1R_2R_5g_m+4C_5R_1R_5+C_5R_2R_5} \\ \text{wo: } & \frac{\sqrt{2R_1R_2g_m+4R_1+R_2+R_5}}{2\sqrt{C_2}\sqrt{C_5}\sqrt{R_1}\sqrt{R_2}\sqrt{R_5}} \\ \text{bandwidth: } & \frac{4C_2R_1R_2+C_2R_2R_5+2C_5R_1R_2R_5g_m+4C_5R_1R_5+C_5R_2R_5}{4C_2C_5R_1R_2R_5} \\ \text{K-LP: } & \frac{R_1R_2R_5g_m-R_1R_2+R_1R_5}{2R_1R_2g_m+4R_1+R_2+R_5} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_2R_1R_2R_5-C_5R_1R_2R_5}{4C_2R_1R_2+C_2R_2R_5+2C_5R_1R_2R_5g_m+4C_5R_1R_5+C_5R_2R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.4 X-INVALID-NUMER-4 $Z(s) = \left(\frac{1}{C_1 s}, R_2, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_2 R_5 s + R_2 R_5 g_m - R_2 + R_5}{C_1 C_5 R_2 R_5 s^2 + 2 R_2 g_m + s (C_1 R_2 + C_1 R_5 + 2 C_5 R_2 R_5 g_m + 4 C_5 R_5) + 4}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_5}\sqrt{R_2}\sqrt{R_5}\sqrt{R_2g_m+2}}{C_1R_2+C_1R_5+2C_5R_2R_5g_m+4C_5R_5} \\ \text{wo: } & \frac{\sqrt{2R_2g_m+4}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_2}\sqrt{R_5}} \\ \text{bandwidth: } & \frac{\sqrt{2}\sqrt{2R_2g_m+4}(C_1R_2+C_1R_5+2C_5R_2R_5g_m+4C_5R_5)}{2C_1C_5R_2R_5\sqrt{R_2g_m+2}} \\ \text{K-LP: } & \frac{R_2R_5g_m-R_2+R_5}{2R_2g_m+4} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5R_2R_5}{C_1R_2+C_1R_5+2C_5R_2R_5g_m+4C_5R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.5 X-INVALID-NUMER-5 $Z(s) = \left(\frac{1}{C_1 s}, \frac{1}{C_2 s}, \infty, \infty, R_5, \infty \right)$

$$H(s) = \frac{C_2 R_5 s + R_5 g_m - 1}{C_1 C_2 R_5 s^2 + 2 g_m + s (C_1 + 4 C_2)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_2}\sqrt{R_5}\sqrt{g_m}}{C_1+4C_2} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1}\sqrt{C_2}\sqrt{R_5}} \\ \text{bandwidth: } & \frac{C_1+4C_2}{C_1C_2R_5} \\ \text{K-LP: } & \frac{R_5g_m-1}{2g_m} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_2R_5}{C_1+4C_2} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.6 X-INVALID-NUMER-6 $Z(s) = \left(\frac{1}{C_1 s}, \frac{1}{C_2 s}, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{R_5 g_m + s (C_2 R_5 - C_5 R_5) - 1}{2 g_m + s^2 (C_1 C_2 R_5 + C_1 C_5 R_5 + 4 C_2 C_5 R_5) + s (C_1 + 4 C_2 + 2 C_5 R_5 g_m)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{R_5}\sqrt{g_m}\sqrt{C_1C_2+C_1C_5+4C_2C_5}}{C_1+4C_2+2C_5R_5g_m} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1C_2R_5+C_1C_5R_5+4C_2C_5R_5}} \end{aligned}$$

bandwidth: $\frac{C_1+4C_2+2C_5R_5g_m}{\sqrt{R_5}\sqrt{C_1C_2+C_1C_5+4C_2C_5}\sqrt{C_1C_2R_5+C_1C_5R_5+4C_2C_5R_5}}$
K-LP: $\frac{R_5g_m-1}{2g_m}$
K-HP: 0
K-BP: $\frac{C_2R_5-C_5R_5}{C_1+4C_2+2C_5R_5g_m}$
Qz: None
Wz: None

8.7 X-INVALID-NUMER-7 $Z(s) = \left(\frac{1}{C_1s}, \frac{R_2}{C_2R_2s+1}, \infty, \infty, R_5, \infty \right)$

$$H(s) = \frac{C_2R_2R_5s + R_2R_5g_m - R_2 + R_5}{C_1C_2R_2R_5s^2 + 2R_2g_m + s(C_1R_2 + C_1R_5 + 4C_2R_2) + 4}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_2}\sqrt{R_2}\sqrt{R_5}\sqrt{R_2g_m+2}}{C_1R_2+C_1R_5+4C_2R_2}$
wo: $\frac{\sqrt{2R_2g_m+4}}{\sqrt{C_1}\sqrt{C_2}\sqrt{R_2}\sqrt{R_5}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2R_2g_m+4}(C_1R_2+C_1R_5+4C_2R_2)}{2C_1C_2R_2R_5\sqrt{R_2g_m+2}}$
K-LP: $\frac{R_2R_5g_m-R_2+R_5}{2R_2g_m+4}$
K-HP: 0
K-BP: $\frac{C_2R_2R_5}{C_1R_2+C_1R_5+4C_2R_2}$
Qz: None
Wz: None

8.8 X-INVALID-NUMER-8 $Z(s) = \left(\frac{1}{C_1s}, \frac{R_2}{C_2R_2s+1}, \infty, \infty, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{R_2R_5g_m - R_2 + R_5 + s(C_2R_2R_5 - C_5R_2R_5)}{2R_2g_m + s^2(C_1C_2R_2R_5 + C_1C_5R_2R_5 + 4C_2C_5R_2R_5) + s(C_1R_2 + C_1R_5 + 4C_2R_2 + 2C_5R_2R_5g_m + 4C_5R_5) + 4}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{R_2}\sqrt{R_5}\sqrt{R_2g_m+2}\sqrt{C_1C_2+C_1C_5+4C_2C_5}}{C_1R_2+C_1R_5+4C_2R_2+2C_5R_2R_5g_m+4C_5R_5}$
wo: $\frac{\sqrt{2R_2g_m+4}}{\sqrt{C_1C_2R_2R_5+C_1C_5R_2R_5+4C_2C_5R_2R_5}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2R_2g_m+4}(C_1R_2+C_1R_5+4C_2R_2+2C_5R_2R_5g_m+4C_5R_5)}{2\sqrt{R_2}\sqrt{R_5}\sqrt{R_2g_m+2}\sqrt{C_1C_2+C_1C_5+4C_2C_5}\sqrt{C_1C_2R_2R_5+C_1C_5R_2R_5+4C_2C_5R_2R_5}}$
K-LP: $\frac{R_2R_5g_m-R_2+R_5}{2R_2g_m+4}$
K-HP: 0
K-BP: $\frac{C_2R_2R_5-C_5R_2R_5}{C_1R_2+C_1R_5+4C_2R_2+2C_5R_2R_5g_m+4C_5R_5}$
Qz: None
Wz: None

8.9 X-INVALID-NUMER-9 $Z(s) = \left(\frac{1}{C_1s}, R_2 + \frac{1}{C_2s}, \infty, \infty, R_5, \infty \right)$

$$H(s) = \frac{R_5g_m + s(C_2R_2R_5g_m - C_2R_2 + C_2R_5) - 1}{2g_m + s^2(C_1C_2R_2 + C_1C_2R_5) + s(C_1 + 2C_2R_2g_m + 4C_2)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_2}\sqrt{g_m}\sqrt{R_2+R_5}}{C_1+2C_2R_2g_m+4C_2}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1C_2R_2+C_1C_2R_5}}$
bandwidth: $\frac{C_1+2C_2R_2g_m+4C_2}{\sqrt{C_1}\sqrt{C_2}\sqrt{R_2+R_5}\sqrt{C_1C_2R_2+C_1C_2R_5}}$
K-LP: $\frac{R_5g_m-1}{2g_m}$
K-HP: 0
K-BP: $\frac{C_2R_2R_5g_m-C_2R_2+C_2R_5}{C_1+2C_2R_2g_m+4C_2}$
Qz: None
Wz: None

8.10 X-INVALID-NUMER-10 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, R_2, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_2 s + R_1 R_2 g_m + R_1}{C_1 C_5 R_1 R_2 s^2 + s(C_1 R_1 + 2C_5 R_1 R_2 g_m + 4C_5 R_1 + C_5 R_2) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_2}}{C_1 R_1 + 2C_5 R_1 R_2 g_m + 4C_5 R_1 + C_5 R_2}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_2}}$
 bandwidth: $\frac{C_1 R_1 + 2C_5 R_1 R_2 g_m + 4C_5 R_1 + C_5 R_2}{C_1 C_5 R_1 R_2}$
 K-LP: $R_1 R_2 g_m + R_1$
 K-HP: 0
 K-BP: $-\frac{C_5 R_1 R_2}{C_1 R_1 + 2C_5 R_1 R_2 g_m + 4C_5 R_1 + C_5 R_2}$
 Qz: None
 Wz: None

8.11 X-INVALID-NUMER-11 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, R_2, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_2 R_5 s + R_1 R_2 R_5 g_m - R_1 R_2 + R_1 R_5}{C_1 C_5 R_1 R_2 R_5 s^2 + 2R_1 R_2 g_m + 4R_1 + R_2 + R_5 + s(C_1 R_1 R_2 + C_1 R_1 R_5 + 2C_5 R_1 R_2 R_5 g_m + 4C_5 R_1 R_5 + C_5 R_2 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_2} \sqrt{R_5} \sqrt{2R_1 R_2 g_m + 4R_1 + R_2 + R_5}}{C_1 R_1 R_2 + C_1 R_1 R_5 + 2C_5 R_1 R_2 R_5 g_m + 4C_5 R_1 R_5 + C_5 R_2 R_5}$
 wo: $\frac{\sqrt{2R_1 R_2 g_m + 4R_1 + R_2 + R_5}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_2} \sqrt{R_5}}$
 bandwidth: $\frac{C_1 R_1 R_2 + C_1 R_1 R_5 + 2C_5 R_1 R_2 R_5 g_m + 4C_5 R_1 R_5 + C_5 R_2 R_5}{C_1 C_5 R_1 R_2 R_5}$
 K-LP: $\frac{R_1 R_2 R_5 g_m - R_1 R_2 + R_1 R_5}{2R_1 R_2 g_m + 4R_1 + R_2 + R_5}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_1 R_2 R_5}{C_1 R_1 R_2 + C_1 R_1 R_5 + 2C_5 R_1 R_2 R_5 g_m + 4C_5 R_1 R_5 + C_5 R_2 R_5}$
 Qz: None
 Wz: None

8.12 X-INVALID-NUMER-12 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, R_2, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_2 g_m + R_1 + s(C_5 R_1 R_2 R_5 g_m - C_5 R_1 R_2 + C_5 R_1 R_5)}{s^2(C_1 C_5 R_1 R_2 + C_1 C_5 R_1 R_5) + s(C_1 R_1 + 2C_5 R_1 R_2 g_m + 4C_5 R_1 + C_5 R_2 + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_2 + R_5}}{C_1 R_1 + 2C_5 R_1 R_2 g_m + 4C_5 R_1 + C_5 R_2 + C_5 R_5}$
 wo: $\frac{1}{\sqrt{C_1 C_5 R_1 R_2 + C_1 C_5 R_1 R_5}}$
 bandwidth: $\frac{C_1 R_1 + 2C_5 R_1 R_2 g_m + 4C_5 R_1 + C_5 R_2 + C_5 R_5}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_2 + R_5} \sqrt{C_1 C_5 R_1 R_2 + C_1 C_5 R_1 R_5}}$
 K-LP: $R_1 R_2 g_m + R_1$
 K-HP: 0
 K-BP: $\frac{C_5 R_1 R_2 R_5 g_m - C_5 R_1 R_2 + C_5 R_1 R_5}{C_1 R_1 + 2C_5 R_1 R_2 g_m + 4C_5 R_1 + C_5 R_2 + C_5 R_5}$
 Qz: None
 Wz: None

8.13 X-INVALID-NUMER-13 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{1}{C_2 s}, \infty, \infty, R_5, \infty \right)$

$$H(s) = \frac{C_2 R_1 R_5 s + R_1 R_5 g_m - R_1}{C_1 C_2 R_1 R_5 s^2 + 2R_1 g_m + s(C_1 R_1 + 4C_2 R_1 + C_2 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{R_5} \sqrt{2R_1 g_m + 1}}{C_1 R_1 + 4C_2 R_1 + C_2 R_5}$
 wo: $\frac{\sqrt{2R_1 g_m + 1}}{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{R_5}}$
 bandwidth: $\frac{C_1 R_1 + 4C_2 R_1 + C_2 R_5}{C_1 C_2 R_1 R_5}$

K-LP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
K-HP: 0
K-BP: $\frac{C_2 R_1 R_5}{C_1 R_1 + 4C_2 R_1 + C_2 R_5}$
Qz: None
Wz: None

8.14 X-INVALID-NUMER-14 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{1}{C_2 s}, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s (C_2 R_1 R_5 - C_5 R_1 R_5)}{2R_1 g_m + s^2 (C_1 C_2 R_1 R_5 + C_1 C_5 R_1 R_5 + 4C_2 C_5 R_1 R_5) + s (C_1 R_1 + 4C_2 R_1 + C_2 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_5} \sqrt{2R_1 g_m + 1} \sqrt{C_1 C_2 + C_1 C_5 + 4C_2 C_5}}{C_1 R_1 + 4C_2 R_1 + C_2 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}$
wo: $\frac{\sqrt{2R_1 g_m + 1}}{\sqrt{C_1 C_2 R_1 R_5 + C_1 C_5 R_1 R_5 + 4C_2 C_5 R_1 R_5}}$
bandwidth: $\frac{C_1 R_1 + 4C_2 R_1 + C_2 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}{\sqrt{R_1} \sqrt{R_5} \sqrt{C_1 C_2 + C_1 C_5 + 4C_2 C_5} \sqrt{C_1 C_2 R_1 R_5 + C_1 C_5 R_1 R_5 + 4C_2 C_5 R_1 R_5}}$
K-LP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
K-HP: 0
K-BP: $\frac{C_2 R_1 R_5 - C_5 R_1 R_5}{C_1 R_1 + 4C_2 R_1 + C_2 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}$
Qz: None
Wz: None

8.15 X-INVALID-NUMER-15 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, R_5, \infty \right)$

$$H(s) = \frac{C_2 R_1 R_2 R_5 s + R_1 R_2 R_5 g_m - R_1 R_2 + R_1 R_5}{C_1 C_2 R_1 R_2 R_5 s^2 + 2R_1 R_2 g_m + 4R_1 + R_2 + R_5 + s (C_1 R_1 R_2 + C_1 R_1 R_5 + 4C_2 R_1 R_2 + C_2 R_2 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{R_2} \sqrt{R_5} \sqrt{2R_1 R_2 g_m + 4R_1 + R_2 + R_5}}{C_1 R_1 R_2 + C_1 R_1 R_5 + 4C_2 R_1 R_2 + C_2 R_2 R_5}$
wo: $\frac{\sqrt{2R_1 R_2 g_m + 4R_1 + R_2 + R_5}}{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{R_2} \sqrt{R_5}}$
bandwidth: $\frac{C_1 R_1 R_2 + C_1 R_1 R_5 + 4C_2 R_1 R_2 + C_2 R_2 R_5}{C_1 C_2 R_1 R_2 R_5}$
K-LP: $\frac{R_1 R_2 R_5 g_m - R_1 R_2 + R_1 R_5}{2R_1 R_2 g_m + 4R_1 + R_2 + R_5}$
K-HP: 0
K-BP: $\frac{C_2 R_1 R_2 R_5}{C_1 R_1 R_2 + C_1 R_1 R_5 + 4C_2 R_1 R_2 + C_2 R_2 R_5}$
Qz: None
Wz: None

8.16 X-INVALID-NUMER-16 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_2 g_m + R_1 + s (C_2 R_1 R_2 - C_5 R_1 R_2)}{s^2 (C_1 C_2 R_1 R_2 + C_1 C_5 R_1 R_2 + 4C_2 C_5 R_1 R_2) + s (C_1 R_1 + C_2 R_2 + 2C_5 R_1 R_2 g_m + 4C_5 R_1 + C_5 R_2) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_2} \sqrt{C_1 C_2 + C_1 C_5 + 4C_2 C_5}}{C_1 R_1 + C_2 R_2 + 2C_5 R_1 R_2 g_m + 4C_5 R_1 + C_5 R_2}$
wo: $\frac{1}{\sqrt{C_1 C_2 R_1 R_2 + C_1 C_5 R_1 R_2 + 4C_2 C_5 R_1 R_2}}$
bandwidth: $\frac{C_1 R_1 + C_2 R_2 + 2C_5 R_1 R_2 g_m + 4C_5 R_1 + C_5 R_2}{\sqrt{R_1} \sqrt{R_2} \sqrt{C_1 C_2 + C_1 C_5 + 4C_2 C_5} \sqrt{C_1 C_2 R_1 R_2 + C_1 C_5 R_1 R_2 + 4C_2 C_5 R_1 R_2}}$
K-LP: $R_1 R_2 g_m + R_1$
K-HP: 0
K-BP: $\frac{C_2 R_1 R_2 - C_5 R_1 R_2}{C_1 R_1 + C_2 R_2 + 2C_5 R_1 R_2 g_m + 4C_5 R_1 + C_5 R_2}$
Qz: None
Wz: None

8.17 X-INVALID-NUMER-17 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{R_1 R_2 R_5 g_m - R_1 R_2 + R_1 R_5 + s (C_2 R_1 R_2 R_5 - C_5 R_1 R_2 R_5)}{2 R_1 R_2 g_m + 4 R_1 + R_2 + R_5 + s^2 (C_1 C_2 R_1 R_2 R_5 + C_1 C_5 R_1 R_2 R_5 + 4 C_2 C_5 R_1 R_2 R_5) + s (C_1 R_1 R_2 + C_1 R_1 R_5 + 4 C_2 R_1 R_2 + C_2 R_2 R_5 + 2 C_5 R_1 R_2 R_5 g_m + 4 C_5 R_1 R_5 + C_5 R_2 R_5)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_2} \sqrt{R_5} \sqrt{C_1 C_2 + C_1 C_5 + 4 C_2 C_5} \sqrt{2 R_1 R_2 g_m + 4 R_1 + R_2 + R_5}}{C_1 R_1 R_2 + C_1 R_1 R_5 + 4 C_2 R_1 R_2 + C_2 R_2 R_5 + 2 C_5 R_1 R_2 R_5 g_m + 4 C_5 R_1 R_5 + C_5 R_2 R_5}$
wo: $\frac{\sqrt{2 R_1 R_2 g_m + 4 R_1 + R_2 + R_5}}{\sqrt{C_1 C_2 R_1 R_2 R_5 + C_1 C_5 R_1 R_2 R_5 + 4 C_2 C_5 R_1 R_2 R_5}}$
bandwidth: $\frac{C_1 R_1 R_2 + C_1 R_1 R_5 + 4 C_2 R_1 R_2 + C_2 R_2 R_5 + 2 C_5 R_1 R_2 R_5 g_m + 4 C_5 R_1 R_5 + C_5 R_2 R_5}{\sqrt{R_1} \sqrt{R_2} \sqrt{R_5} \sqrt{C_1 C_2 + C_1 C_5 + 4 C_2 C_5} \sqrt{C_1 C_2 R_1 R_2 R_5 + C_1 C_5 R_1 R_2 R_5 + 4 C_2 C_5 R_1 R_2 R_5}}$
K-LP: $\frac{R_1 R_2 R_5 g_m - R_1 R_2 + R_1 R_5}{2 R_1 R_2 g_m + 4 R_1 + R_2 + R_5}$
K-HP: 0
K-BP: $\frac{C_2 R_1 R_2 R_5 - C_5 R_1 R_2 R_5}{C_1 R_1 R_2 + C_1 R_1 R_5 + 4 C_2 R_1 R_2 + C_2 R_2 R_5 + 2 C_5 R_1 R_2 R_5 g_m + 4 C_5 R_1 R_5 + C_5 R_2 R_5}$
Qz: None
Wz: None

8.18 X-INVALID-NUMER-18 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, R_2 + \frac{1}{C_2 s}, \infty, \infty, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s (C_2 R_1 R_2 R_5 g_m - C_2 R_1 R_2 + C_2 R_1 R_5)}{2 R_1 g_m + s^2 (C_1 C_2 R_1 R_2 + C_1 C_2 R_1 R_5) + s (C_1 R_1 + 2 C_2 R_1 R_2 g_m + 4 C_2 R_1 + C_2 R_2 + C_2 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{R_2 + R_5} \sqrt{2 R_1 g_m + 1}}{C_1 R_1 + 2 C_2 R_1 R_2 g_m + 4 C_2 R_1 + C_2 R_2 + C_2 R_5}$
wo: $\frac{\sqrt{2 R_1 g_m + 1}}{\sqrt{C_1 C_2 R_1 R_2 + C_1 C_2 R_1 R_5}}$
bandwidth: $\frac{C_1 R_1 + 2 C_2 R_1 R_2 g_m + 4 C_2 R_1 + C_2 R_2 + C_2 R_5}{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{R_2 + R_5} \sqrt{C_1 C_2 R_1 R_2 + C_1 C_2 R_1 R_5}}$
K-LP: $\frac{R_1 R_5 g_m - R_1}{2 R_1 g_m + 1}$
K-HP: 0
K-BP: $\frac{C_2 R_1 R_2 R_5 g_m - C_2 R_1 R_2 + C_2 R_1 R_5}{C_1 R_1 + 2 C_2 R_1 R_2 g_m + 4 C_2 R_1 + C_2 R_2 + C_2 R_5}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1, R_2, \infty, \infty, R_5, \infty)$

$$H(s) = \frac{R_1 R_2 R_5 g_m - R_1 R_2 + R_1 R_5}{2 R_1 R_2 g_m + 4 R_1 + R_2 + R_5}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1, R_2, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_2 s + R_1 R_2 g_m + R_1}{s (2 C_5 R_1 R_2 g_m + 4 C_5 R_1 + C_5 R_2) + 1}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1, R_2, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_2 R_5 s + R_1 R_2 R_5 g_m - R_1 R_2 + R_1 R_5}{2 R_1 R_2 g_m + 4 R_1 + R_2 + R_5 + s (2 C_5 R_1 R_2 R_5 g_m + 4 C_5 R_1 R_5 + C_5 R_2 R_5)}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1, R_2, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_2 g_m + R_1 + s (C_5 R_1 R_2 R_5 g_m - C_5 R_1 R_2 + C_5 R_1 R_5)}{s (2 C_5 R_1 R_2 g_m + 4 C_5 R_1 + C_5 R_2 + C_5 R_5) + 1}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(R_1, \frac{1}{C_2 s}, \infty, \infty, R_5, \infty \right)$

$$H(s) = \frac{C_2 R_1 R_5 s + R_1 R_5 g_m - R_1}{2 R_1 g_m + s (4 C_2 R_1 + C_2 R_5) + 1}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1, \frac{1}{C_2 s}, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 g_m + s (C_2 R_1 - C_5 R_1)}{4 C_2 C_5 R_1 s^2 + s (C_2 + 2 C_5 R_1 g_m + C_5)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(R_1, \frac{1}{C_2 s}, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_2 C_5 R_1 R_5 s^2 + R_1 g_m + s (C_2 R_1 + C_5 R_1 R_5 g_m - C_5 R_1)}{s^2 (4 C_2 C_5 R_1 + C_2 C_5 R_5) + s (C_2 + 2 C_5 R_1 g_m + C_5)}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(R_1, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, R_5, \infty \right)$

$$H(s) = \frac{C_2 R_1 R_2 R_5 s + R_1 R_2 R_5 g_m - R_1 R_2 + R_1 R_5}{2 R_1 R_2 g_m + 4 R_1 + R_2 + R_5 + s (4 C_2 R_1 R_2 + C_2 R_2 R_5)}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(R_1, R_2 + \frac{1}{C_2 s}, \infty, \infty, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s (C_2 R_1 R_2 R_5 g_m - C_2 R_1 R_2 + C_2 R_1 R_5)}{2 R_1 g_m + s (2 C_2 R_1 R_2 g_m + 4 C_2 R_1 + C_2 R_2 + C_2 R_5) + 1}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(R_1, R_2 + \frac{1}{C_2 s}, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_2 C_5 R_1 R_2 s^2 + R_1 g_m + s (C_2 R_1 R_2 g_m + C_2 R_1 - C_5 R_1)}{s^2 (2 C_2 C_5 R_1 R_2 g_m + 4 C_2 C_5 R_1 + C_2 C_5 R_2) + s (C_2 + 2 C_5 R_1 g_m + C_5)}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(R_1, R_2 + \frac{1}{C_2 s}, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 g_m + s^2 (C_2 C_5 R_1 R_2 R_5 g_m - C_2 C_5 R_1 R_2 + C_2 C_5 R_1 R_5) + s (C_2 R_1 R_2 g_m + C_2 R_1 + C_5 R_1 R_5 g_m - C_5 R_1)}{s^2 (2 C_2 C_5 R_1 R_2 g_m + 4 C_2 C_5 R_1 + C_2 C_5 R_2 + C_2 C_5 R_5) + s (C_2 + 2 C_5 R_1 g_m + C_5)}$$

9.12 X-INVALID-ORDER-12 $Z(s) = \left(\frac{1}{C_1 s}, R_2, \infty, \infty, R_5, \infty \right)$

$$H(s) = \frac{R_2 R_5 g_m - R_2 + R_5}{2 R_2 g_m + s (C_1 R_2 + C_1 R_5) + 4}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(\frac{1}{C_1 s}, R_2, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_2 s + R_2 g_m + 1}{C_1 C_5 R_2 s^2 + s (C_1 + 2 C_5 R_2 g_m + 4 C_5)}$$

9.14 X-INVALID-ORDER-14 $Z(s) = \left(\frac{1}{C_1 s}, R_2, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_2 g_m + s (C_5 R_2 R_5 g_m - C_5 R_2 + C_5 R_5) + 1}{s^2 (C_1 C_5 R_2 + C_1 C_5 R_5) + s (C_1 + 2 C_5 R_2 g_m + 4 C_5)}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(\frac{1}{C_1 s}, \frac{1}{C_2 s}, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{g_m + s(C_2 - C_5)}{2C_5 g_m s + s^2(C_1 C_2 + C_1 C_5 + 4C_2 C_5)}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(\frac{1}{C_1 s}, \frac{1}{C_2 s}, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_2 C_5 R_5 s^2 + g_m + s(C_2 + C_5 R_5 g_m - C_5)}{C_1 C_2 C_5 R_5 s^3 + 2C_5 g_m s + s^2(C_1 C_2 + C_1 C_5 + 4C_2 C_5)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(\frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_2 g_m + s(C_2 R_2 - C_5 R_2) + 1}{s^2(C_1 C_2 R_2 + C_1 C_5 R_2 + 4C_2 C_5 R_2) + s(C_1 + 2C_5 R_2 g_m + 4C_5)}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_2 C_5 R_2 R_5 s^2 + R_2 g_m + s(C_2 R_2 + C_5 R_2 R_5 g_m - C_5 R_2 + C_5 R_5) + 1}{C_1 C_2 C_5 R_2 R_5 s^3 + s^2(C_1 C_2 R_2 + C_1 C_5 R_2 + C_1 C_5 R_5 + 4C_2 C_5 R_2) + s(C_1 + 2C_5 R_2 g_m + 4C_5)}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_2 C_5 R_2 s^2 + g_m + s(C_2 R_2 g_m + C_2 - C_5)}{C_1 C_2 C_5 R_2 s^3 + 2C_5 g_m s + s^2(C_1 C_2 + C_1 C_5 + 2C_2 C_5 R_2 g_m + 4C_2 C_5)}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_2 C_5 R_2 R_5 s^2 + R_5 g_m + s(C_2 R_2 R_5 g_m - C_2 R_2 + C_2 R_5 - C_5 R_5) - 1}{C_1 C_2 C_5 R_2 R_5 s^3 + 2g_m + s^2(C_1 C_2 R_2 + C_1 C_2 R_5 + C_1 C_5 R_5 + 2C_2 C_5 R_2 R_5 g_m + 4C_2 C_5 R_5) + s(C_1 + 2C_2 R_2 g_m + 4C_2 + 2C_5 R_5 g_m)}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{g_m + s^2(C_2 C_5 R_2 R_5 g_m - C_2 C_5 R_2 + C_2 C_5 R_5) + s(C_2 R_2 g_m + C_2 + C_5 R_5 g_m - C_5)}{2C_5 g_m s + s^3(C_1 C_2 C_5 R_2 + C_1 C_2 C_5 R_5) + s^2(C_1 C_2 + C_1 C_5 + 2C_2 C_5 R_2 g_m + 4C_2 C_5)}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, R_2, \infty, \infty, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_2 R_5 g_m - R_1 R_2 + R_1 R_5}{2R_1 R_2 g_m + 4R_1 + R_2 + R_5 + s(C_1 R_1 R_2 + C_1 R_1 R_5)}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{1}{C_2 s}, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s(C_2 R_1 - C_5 R_1)}{s^2(C_1 C_2 R_1 + C_1 C_5 R_1 + 4C_2 C_5 R_1) + s(C_2 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{1}{C_2 s}, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_2 C_5 R_1 R_5 s^2 + R_1 g_m + s(C_2 R_1 + C_5 R_1 R_5 g_m - C_5 R_1)}{C_1 C_2 C_5 R_1 R_5 s^3 + s^2(C_1 C_2 R_1 + C_1 C_5 R_1 + 4C_2 C_5 R_1 + C_2 C_5 R_5) + s(C_2 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_2 C_5 R_1 R_2 R_5 s^2 + R_1 R_2 g_m + R_1 + s(C_2 R_1 R_2 + C_5 R_1 R_2 R_5 g_m - C_5 R_1 R_2 + C_5 R_1 R_5)}{C_1 C_2 C_5 R_1 R_2 R_5 s^3 + s^2(C_1 C_2 R_1 R_2 + C_1 C_5 R_1 R_2 + C_1 C_5 R_1 R_5 + 4C_2 C_5 R_1 R_2 + C_2 C_5 R_2 R_5) + s(C_1 R_1 + C_2 R_2 + 2C_5 R_1 R_2 g_m + 4C_5 R_1 + C_5 R_2 + C_5 R_5) + 1}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_2 C_5 R_1 R_2 s^2 + R_1 g_m + s(C_2 R_1 R_2 g_m + C_2 R_1 - C_5 R_1)}{C_1 C_2 C_5 R_1 R_2 s^3 + s^2(C_1 C_2 R_1 + C_1 C_5 R_1 + 2C_2 C_5 R_1 R_2 g_m + 4C_2 C_5 R_1 + C_2 C_5 R_2) + s(C_2 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.27 \quad X-INVALID-ORDER-27} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_2 C_5 R_1 R_2 R_5 s^2 + R_1 R_5 g_m - R_1 + s(C_2 R_1 R_2 R_5 g_m - C_2 R_1 R_2 + C_2 R_1 R_5 - C_5 R_1 R_5)}{C_1 C_2 C_5 R_1 R_2 R_5 s^3 + 2R_1 g_m + s^2(C_1 C_2 R_1 R_2 + C_1 C_2 R_1 R_5 + C_1 C_5 R_1 R_5 + 2C_2 C_5 R_1 R_2 R_5 g_m + 4C_2 C_5 R_1 R_5 + C_2 C_5 R_2 R_5) + s(C_1 R_1 + 2C_2 R_1 R_2 g_m + 4C_2 R_1 + C_2 R_2 + C_2 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

$$\mathbf{9.28 \quad X-INVALID-ORDER-28} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, R_2 + \frac{1}{C_2 s}, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^2(C_2 C_5 R_1 R_2 R_5 g_m - C_2 C_5 R_1 R_2 + C_2 C_5 R_1 R_5) + s(C_2 R_1 R_2 g_m + C_2 R_1 + C_5 R_1 R_5 g_m - C_5 R_1)}{s^3(C_1 C_2 C_5 R_1 R_2 + C_1 C_2 C_5 R_1 R_5) + s^2(C_1 C_2 R_1 + C_1 C_5 R_1 + 2C_2 C_5 R_1 R_2 g_m + 4C_2 C_5 R_1 + C_2 C_5 R_2 + C_2 C_5 R_5) + s(C_2 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.29 \quad X-INVALID-ORDER-29} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2, \infty, \infty, R_5, \infty \right)$$

$$H(s) = \frac{R_2 R_5 g_m - R_2 + R_5 + s(C_1 R_1 R_2 R_5 g_m - C_1 R_1 R_2 + C_1 R_1 R_5)}{2R_2 g_m + s(2C_1 R_1 R_2 g_m + 4C_1 R_1 + C_1 R_2 + C_1 R_5) + 4}$$

$$\mathbf{9.30 \quad X-INVALID-ORDER-30} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_2 s^2 + R_2 g_m + s(C_1 R_1 R_2 g_m + C_1 R_1 - C_5 R_2) + 1}{s^2(2C_1 C_5 R_1 R_2 g_m + 4C_1 C_5 R_1 + C_1 C_5 R_2) + s(C_1 + 2C_5 R_2 g_m + 4C_5)}$$

$$\mathbf{9.31 \quad X-INVALID-ORDER-31} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_2 g_m + s^2(C_1 C_5 R_1 R_2 R_5 g_m - C_1 C_5 R_1 R_2 + C_1 C_5 R_1 R_5) + s(C_1 R_1 R_2 g_m + C_1 R_1 + C_5 R_2 R_5 g_m - C_5 R_2 + C_5 R_5) + 1}{s^2(2C_1 C_5 R_1 R_2 g_m + 4C_1 C_5 R_1 + C_1 C_5 R_2 + C_1 C_5 R_5) + s(C_1 + 2C_5 R_2 g_m + 4C_5)}$$

$$\mathbf{9.32 \quad X-INVALID-ORDER-32} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{1}{C_2 s}, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{g_m + s^2(C_1 C_2 R_1 - C_1 C_5 R_1) + s(C_1 R_1 g_m + C_2 - C_5)}{4C_1 C_2 C_5 R_1 s^3 + 2C_5 g_m s + s^2(C_1 C_2 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 4C_2 C_5)}$$

$$\mathbf{9.33 \quad X-INVALID-ORDER-33} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{1}{C_2 s}, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^2(C_1 C_2 R_1 R_5 - C_1 C_5 R_1 R_5) + s(C_1 R_1 R_5 g_m - C_1 R_1 + C_2 R_5 - C_5 R_5) - 1}{4C_1 C_2 C_5 R_1 R_5 s^3 + 2g_m + s^2(4C_1 C_2 R_1 + C_1 C_2 R_5 + 2C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5 + 4C_2 C_5 R_5) + s(2C_1 R_1 g_m + C_1 + 4C_2 + 2C_5 R_5 g_m)}$$

$$\mathbf{9.34 \quad X-INVALID-ORDER-34} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{1}{C_2 s}, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_1 C_2 C_5 R_1 R_5 s^3 + g_m + s^2(C_1 C_2 R_1 + C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_2 C_5 R_5) + s(C_1 R_1 g_m + C_2 + C_5 R_5 g_m - C_5)}{2C_5 g_m s + s^3(4C_1 C_2 C_5 R_1 + C_1 C_2 C_5 R_5) + s^2(C_1 C_2 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 4C_2 C_5)}$$

9.35 X-INVALID-ORDER-35 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_2 g_m + s^2 (C_1 C_2 R_1 R_2 - C_1 C_5 R_1 R_2) + s (C_1 R_1 R_2 g_m + C_1 R_1 + C_2 R_2 - C_5 R_2) + 1}{4 C_1 C_2 C_5 R_1 R_2 s^3 + s^2 (C_1 C_2 R_2 + 2 C_1 C_5 R_1 R_2 g_m + 4 C_1 C_5 R_1 + C_1 C_5 R_2 + 4 C_2 C_5 R_2) + s (C_1 + 2 C_5 R_2 g_m + 4 C_5)}$$

9.36 X-INVALID-ORDER-36 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{R_2 R_5 g_m - R_2 + R_5 + s^2 (C_1 C_2 R_1 R_2 R_5 - C_1 C_5 R_1 R_2 R_5) + s (C_1 R_1 R_2 R_5 g_m - C_1 R_1 R_2 + C_1 R_1 R_5 + C_2 R_2 R_5 - C_5 R_2 R_5)}{4 C_1 C_2 C_5 R_1 R_2 R_5 s^3 + 2 R_2 g_m + s^2 (4 C_1 C_2 R_1 R_2 + C_1 C_2 R_2 R_5 + 2 C_1 C_5 R_1 R_2 R_5 g_m + 4 C_1 C_5 R_1 R_5 + C_1 C_5 R_2 R_5 + 4 C_2 C_5 R_2 R_5) + s (2 C_1 R_1 R_2 g_m + 4 C_1 R_1 + C_1 R_2 + C_1 R_5 + 4 C_2 R_2 + 2 C_5 R_2 R_5 g_m + 4 C_5 R_5) + 4}$$

9.37 X-INVALID-ORDER-37 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_2 C_5 R_1 R_2 R_5 s^3 + R_2 g_m + s^2 (C_1 C_2 R_1 R_2 + C_1 C_5 R_1 R_2 R_5 g_m - C_1 C_5 R_1 R_2 + C_1 C_5 R_1 R_5 + C_2 C_5 R_2 R_5) + s (C_1 R_1 R_2 g_m + C_1 R_1 + C_2 R_2 + C_5 R_2 R_5 g_m - C_5 R_2 + C_5 R_5) + 1}{s^3 (4 C_1 C_2 C_5 R_1 R_2 + C_1 C_2 C_5 R_2 R_5) + s^2 (C_1 C_2 R_2 + 2 C_1 C_5 R_1 R_2 g_m + 4 C_1 C_5 R_1 + C_1 C_5 R_2 + C_1 C_5 R_5 + 4 C_2 C_5 R_2) + s (C_1 + 2 C_5 R_2 g_m + 4 C_5)}$$

9.38 X-INVALID-ORDER-38 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_2 C_5 R_1 R_2 s^3 + g_m + s^2 (C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1 - C_1 C_5 R_1 - C_2 C_5 R_2) + s (C_1 R_1 g_m + C_2 R_2 g_m + C_2 - C_5)}{2 C_5 g_m s + s^3 (2 C_1 C_2 C_5 R_1 R_2 g_m + 4 C_1 C_2 C_5 R_1 + C_1 C_2 C_5 R_2) + s^2 (C_1 C_2 + 2 C_1 C_5 R_1 g_m + C_1 C_5 + 2 C_2 C_5 R_2 g_m + 4 C_2 C_5)}$$

9.39 X-INVALID-ORDER-39 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_2 C_5 R_1 R_2 R_5 s^3 + R_5 g_m + s^2 (C_1 C_2 R_1 R_2 R_5 g_m - C_1 C_2 R_1 R_2 + C_1 C_2 R_1 R_5 - C_1 C_5 R_1 R_5 - C_2 C_5 R_2 R_5) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_2 R_2 R_5 g_m - C_2 R_2 + C_2 R_5 - C_5 R_5) - 1}{2 g_m + s^3 (2 C_1 C_2 C_5 R_1 R_2 R_5 g_m + 4 C_1 C_2 C_5 R_1 R_5 + C_1 C_2 C_5 R_2 R_5) + s^2 (2 C_1 C_2 R_1 R_2 g_m + 4 C_1 C_2 R_1 + C_1 C_2 R_2 + C_1 C_2 R_5 + 2 C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5 + 2 C_2 C_5 R_2 R_5 g_m + 4 C_2 C_5 R_5) + s (2 C_1 R_1 g_m + C_1 + 2 C_2 R_2 g_m + 4 C_2 + 2 C_5 R_5 g_m)}$$

9.40 X-INVALID-ORDER-40 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{g_m + s^3 (C_1 C_2 C_5 R_1 R_2 R_5 g_m - C_1 C_2 C_5 R_1 R_2 + C_1 C_2 C_5 R_1 R_5) + s^2 (C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1 + C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_2 C_5 R_2 R_5 g_m - C_2 C_5 R_2 + C_2 C_5 R_5) + s (C_1 R_1 g_m + C_2 R_2 g_m + C_2 + C_5 R_5 g_m - C_5)}{2 C_5 g_m s + s^3 (2 C_1 C_2 C_5 R_1 R_2 g_m + 4 C_1 C_2 C_5 R_1 + C_1 C_2 C_5 R_2 + C_1 C_2 C_5 R_5) + s^2 (C_1 C_2 + 2 C_1 C_5 R_1 g_m + C_1 C_5 + 2 C_2 C_5 R_2 g_m + 4 C_2 C_5)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(R_1, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_2 C_5 R_1 R_2 R_5 s^2 + R_1 R_2 g_m + R_1 + s (C_2 R_1 R_2 + C_5 R_1 R_2 R_5 g_m - C_5 R_1 R_2 + C_5 R_1 R_5)}{s^2 (4 C_2 C_5 R_1 R_2 + C_2 C_5 R_2 R_5) + s (C_2 R_2 + 2 C_5 R_1 R_2 g_m + 4 C_5 R_1 + C_5 R_2 + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_2} \sqrt{C_5} \sqrt{R_2} \sqrt{4 R_1 + R_5}}{C_2 R_2 + 2 C_5 R_1 R_2 g_m + 4 C_5 R_1 + C_5 R_2 + C_5 R_5}$
 wo: $\frac{1}{\sqrt{4 C_2 C_5 R_1 R_2 + C_2 C_5 R_2 R_5}}$
 bandwidth: $\frac{C_2 R_2 + 2 C_5 R_1 R_2 g_m + 4 C_5 R_1 + C_5 R_2 + C_5 R_5}{\sqrt{C_2} \sqrt{C_5} \sqrt{R_2} \sqrt{4 R_1 + R_5} \sqrt{4 C_2 C_5 R_1 R_2 + C_2 C_5 R_2 R_5}}$
 K-LP: $R_1 R_2 g_m + R_1$
 K-HP: $\frac{R_1 R_5}{4 R_1 + R_5}$
 K-BP: $\frac{C_2 R_1 R_2 + C_5 R_1 R_2 R_5 g_m - C_5 R_1 R_2 + C_5 R_1 R_5}{C_2 R_2 + 2 C_5 R_1 R_2 g_m + 4 C_5 R_1 + C_5 R_2 + C_5 R_5}$
 Qz: None
 Wz: $\frac{\sqrt{R_2 g_m + 1}}{\sqrt{C_2} \sqrt{C_5} \sqrt{R_2} \sqrt{R_5}}$

10.2 X-INVALID-WZ-2 $Z(s) = \left(R_1, R_2 + \frac{1}{C_2 s}, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_2 C_5 R_1 R_2 R_5 s^2 + R_1 R_5 g_m - R_1 + s (C_2 R_1 R_2 R_5 g_m - C_2 R_1 R_2 + C_2 R_1 R_5 - C_5 R_1 R_5)}{2 R_1 g_m + s^2 (2 C_2 C_5 R_1 R_2 R_5 g_m + 4 C_2 C_5 R_1 R_5 + C_2 C_5 R_2 R_5) + s (2 C_2 R_1 R_2 g_m + 4 C_2 R_1 + C_2 R_2 + C_2 R_5 + 2 C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_2}\sqrt{C_5}R_1R_2\sqrt{R_5}g_m\sqrt{\frac{2R_1g_m}{2R_1R_2g_m+4R_1+R_2}+\frac{1}{2R_1R_2g_m+4R_1+R_2}}+4\sqrt{C_2}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_2g_m+4R_1+R_2}+\frac{1}{2R_1R_2g_m+4R_1+R_2}}+\sqrt{C_2}\sqrt{C_5}R_2\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_2g_m+4R_1+R_2}+\frac{1}{2R_1R_2g_m+4R_1+R_2}}}{2C_2R_1R_2g_m+4C_2R_1+C_2R_2+C_2R_5+2C_5R_1R_5g_m+C_5R_5} \\ \text{wo: } & \sqrt{\frac{2R_1g_m+1}{2C_2C_5R_1R_2R_5g_m+4C_2C_5R_1R_5+C_2C_5R_2R_5}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{2R_1g_m+1}{2C_2C_5R_1R_2R_5g_m+4C_2C_5R_1R_5+C_2C_5R_2R_5}}(2C_2R_1R_2g_m+4C_2R_1+C_2R_2+C_2R_5+2C_5R_1R_5g_m+C_5R_5)}{2\sqrt{C_2}\sqrt{C_5}R_1R_2\sqrt{R_5}g_m\sqrt{\frac{2R_1g_m}{2R_1R_2g_m+4R_1+R_2}+\frac{1}{2R_1R_2g_m+4R_1+R_2}}+4\sqrt{C_2}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_2g_m+4R_1+R_2}+\frac{1}{2R_1R_2g_m+4R_1+R_2}}+\sqrt{C_2}\sqrt{C_5}R_2\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_2g_m+4R_1+R_2}+\frac{1}{2R_1R_2g_m+4R_1+R_2}}} \\ \text{K-LP: } & \frac{R_1R_5g_m-R_1}{2R_1g_m+1} \\ \text{K-HP: } & -\frac{R_1R_2}{2R_1R_2g_m+4R_1+R_2} \\ \text{K-BP: } & \frac{C_2R_1R_2R_5g_m-C_2R_1R_2+C_2R_1R_5-C_5R_1R_5}{2C_2R_1R_2g_m+4C_2R_1+C_2R_2+C_2R_5+2C_5R_1R_5g_m+C_5R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{\sqrt{-R_5g_m+1}}{\sqrt{C_2}\sqrt{C_5}\sqrt{R_2}\sqrt{R_5}} \end{aligned}$$

10.3 X-INVALID-WZ-3 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_2 R_5 s^2 + R_2 R_5 g_m - R_2 + R_5 + s (C_1 R_1 R_2 R_5 g_m - C_1 R_1 R_2 + C_1 R_1 R_5 - C_5 R_2 R_5)}{2 R_2 g_m + s^2 (2 C_1 C_5 R_1 R_2 R_5 g_m + 4 C_1 C_5 R_1 R_5 + C_1 C_5 R_2 R_5) + s (2 C_1 R_1 R_2 g_m + 4 C_1 R_1 + C_1 R_2 + C_1 R_5 + 2 C_5 R_2 R_5 g_m + 4 C_5 R_5) + 4}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_2\sqrt{R_5}g_m\sqrt{\frac{R_2g_m}{2R_1R_2g_m+4R_1+R_2}+\frac{2}{2R_1R_2g_m+4R_1+R_2}}+4\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{R_2g_m}{2R_1R_2g_m+4R_1+R_2}+\frac{2}{2R_1R_2g_m+4R_1+R_2}}+\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_2\sqrt{R_5}\sqrt{\frac{R_2g_m}{2R_1R_2g_m+4R_1+R_2}+\frac{2}{2R_1R_2g_m+4R_1+R_2}}}{2C_1R_1R_2g_m+4C_1R_1+C_1R_2+C_1R_5+2C_5R_2R_5g_m+4C_5R_5} \\ \text{wo: } & \sqrt{\frac{2R_2g_m+4}{2C_1C_5R_1R_2R_5g_m+4C_1C_5R_1R_5+C_1C_5R_2R_5}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{2R_2g_m+4}{2C_1C_5R_1R_2R_5g_m+4C_1C_5R_1R_5+C_1C_5R_2R_5}}(2C_1R_1R_2g_m+4C_1R_1+C_1R_2+C_1R_5+2C_5R_2R_5g_m+4C_5R_5)}{2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_2\sqrt{R_5}g_m\sqrt{\frac{R_2g_m}{2R_1R_2g_m+4R_1+R_2}+\frac{2}{2R_1R_2g_m+4R_1+R_2}}+4\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{R_2g_m}{2R_1R_2g_m+4R_1+R_2}+\frac{2}{2R_1R_2g_m+4R_1+R_2}}+\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_2\sqrt{R_5}\sqrt{\frac{R_2g_m}{2R_1R_2g_m+4R_1+R_2}+\frac{2}{2R_1R_2g_m+4R_1+R_2}}} \\ \text{K-LP: } & \frac{R_2R_5g_m-R_2+R_5}{2R_2g_m+4} \\ \text{K-HP: } & -\frac{R_1R_2}{2R_1R_2g_m+4R_1+R_2} \\ \text{K-BP: } & \frac{C_1R_1R_2R_5g_m-C_1R_1R_2+C_1R_1R_5-C_5R_2R_5}{2C_1R_1R_2g_m+4C_1R_1+C_1R_2+C_1R_5+2C_5R_2R_5g_m+4C_5R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{\sqrt{-R_2R_5g_m+R_2-R_5}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_2}\sqrt{R_5}} \end{aligned}$$

10.4 X-INVALID-WZ-4 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{1}{C_2 s}, \infty, \infty, R_5, \infty \right)$

$$H(s) = \frac{C_1 C_2 R_1 R_5 s^2 + R_5 g_m + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_2 R_5) - 1}{2 g_m + s^2 (4 C_1 C_2 R_1 + C_1 C_2 R_5) + s (2 C_1 R_1 g_m + C_1 + 4 C_2)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_2}\sqrt{g_m}\sqrt{4R_1+R_5}}{2C_1R_1g_m+C_1+4C_2} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{g_m}}{\sqrt{4C_1C_2R_1+C_1C_2R_5}} \\ \text{bandwidth: } & \frac{2C_1R_1g_m+C_1+4C_2}{\sqrt{C_1}\sqrt{C_2}\sqrt{4R_1+R_5}\sqrt{4C_1C_2R_1+C_1C_2R_5}} \\ \text{K-LP: } & \frac{R_5g_m-1}{2g_m} \\ \text{K-HP: } & \frac{R_1R_5}{4R_1+R_5} \\ \text{K-BP: } & \frac{C_1R_1R_5g_m-C_1R_1+C_2R_5}{2C_1R_1g_m+C_1+4C_2} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{\sqrt{R_5g_m-1}}{\sqrt{C_1}\sqrt{C_2}\sqrt{R_1}\sqrt{R_5}} \end{aligned}$$

10.5 X-INVALID-WZ-5 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, R_5, \infty \right)$

$$H(s) = \frac{C_1 C_2 R_1 R_2 R_5 s^2 + R_2 R_5 g_m - R_2 + R_5 + s (C_1 R_1 R_2 R_5 g_m - C_1 R_1 R_2 + C_1 R_1 R_5 + C_2 R_2 R_5)}{2 R_2 g_m + s^2 (4 C_1 C_2 R_1 R_2 + C_1 C_2 R_2 R_5) + s (2 C_1 R_1 R_2 g_m + 4 C_1 R_1 + C_1 R_2 + C_1 R_5 + 4 C_2 R_2) + 4}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_2}\sqrt{R_2}\sqrt{4R_1+R_5}\sqrt{R_2g_m+2}}{2C_1R_1R_2g_m+4C_1R_1+C_1R_2+C_1R_5+4C_2R_2}$
 wo: $\frac{\sqrt{2R_2g_m+4}}{\sqrt{4C_1C_2R_1R_2+C_1C_2R_2R_5}}$
 bandwidth: $\frac{\sqrt{2}\sqrt{2R_2g_m+4}(2C_1R_1R_2g_m+4C_1R_1+C_1R_2+C_1R_5+4C_2R_2)}{2\sqrt{C_1}\sqrt{C_2}\sqrt{R_2}\sqrt{4R_1+R_5}\sqrt{R_2g_m+2}\sqrt{4C_1C_2R_1R_2+C_1C_2R_2R_5}}$
 K-LP: $\frac{R_2R_5g_m-R_2+R_5}{2R_2g_m+4}$
 K-HP: $\frac{R_1R_5}{4R_1+R_5}$
 K-BP: $\frac{C_1R_1R_2R_5g_m-C_1R_1R_2+C_1R_1R_5+C_2R_2R_5}{2C_1R_1R_2g_m+4C_1R_1+C_1R_2+C_1R_5+4C_2R_2}$
 Qz: None
 Wz: $\frac{\sqrt{R_2R_5g_m-R_2+R_5}}{\sqrt{C_1}\sqrt{C_2}\sqrt{R_1}\sqrt{R_2}\sqrt{R_5}}$

10.6 X-INVALID-WZ-6 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, \infty, R_5, \infty \right)$

$$H(s) = \frac{R_5 g_m + s^2 (C_1 C_2 R_1 R_2 R_5 g_m - C_1 C_2 R_1 R_2 + C_1 C_2 R_1 R_5) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_2 R_2 R_5 g_m - C_2 R_2 + C_2 R_5) - 1}{2 g_m + s^2 (2 C_1 C_2 R_1 R_2 g_m + 4 C_1 C_2 R_1 + C_1 C_2 R_2 + C_1 C_2 R_5) + s (2 C_1 R_1 g_m + C_1 + 2 C_2 R_2 g_m + 4 C_2)}$$

Parameters:

Q: $\frac{2\sqrt{2}\sqrt{C_1}\sqrt{C_2}R_1R_2g_m\sqrt{\frac{g_m}{2R_1R_2g_m+4R_1+R_2+R_5}}+4\sqrt{2}\sqrt{C_1}\sqrt{C_2}R_1\sqrt{\frac{g_m}{2R_1R_2g_m+4R_1+R_2+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_2}R_2\sqrt{\frac{g_m}{2R_1R_2g_m+4R_1+R_2+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_2}R_5\sqrt{\frac{g_m}{2R_1R_2g_m+4R_1+R_2+R_5}}}{2C_1R_1g_m+C_1+2C_2R_2g_m+4C_2}$
 wo: $\sqrt{2}\sqrt{\frac{g_m}{2C_1C_2R_1R_2g_m+4C_1C_2R_1+C_1C_2R_2+C_1C_2R_5}}$
 bandwidth: $\frac{\sqrt{2}\sqrt{\frac{g_m}{2C_1C_2R_1R_2g_m+4C_1C_2R_1+C_1C_2R_2+C_1C_2R_5}}(2C_1R_1g_m+C_1+2C_2R_2g_m+4C_2)}{2\sqrt{2}\sqrt{C_1}\sqrt{C_2}R_1R_2g_m\sqrt{\frac{g_m}{2R_1R_2g_m+4R_1+R_2+R_5}}+4\sqrt{2}\sqrt{C_1}\sqrt{C_2}R_1\sqrt{\frac{g_m}{2R_1R_2g_m+4R_1+R_2+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_2}R_2\sqrt{\frac{g_m}{2R_1R_2g_m+4R_1+R_2+R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_2}R_5\sqrt{\frac{g_m}{2R_1R_2g_m+4R_1+R_2+R_5}}}$
 K-LP: $\frac{R_5g_m-1}{2g_m}$
 K-HP: $\frac{R_1R_2R_5g_m-R_1R_2+R_1R_5}{2R_1R_2g_m+4R_1+R_2+R_5}$
 K-BP: $\frac{C_1R_1R_5g_m-C_1R_1+C_2R_2R_5g_m-C_2R_2+C_2R_5}{2C_1R_1g_m+C_1+2C_2R_2g_m+4C_2}$
 Qz: None
 Wz: $\sqrt{\frac{R_5g_m-1}{C_1C_2R_1R_2R_5g_m-C_1C_2R_1R_2+C_1C_2R_1R_5}}$

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